GALAHAD SBLS

USER DOCUMENTATION GALAHAD Optimization Library version 5.5

1 SUMMARY

Given a **block, symmetric matrix**

$$K_H = \begin{pmatrix} H & A^T \\ A & -C \end{pmatrix},$$

this package constructs a variety of **preconditioners** of the form

$$K_G = \begin{pmatrix} G & A^T \\ A & -C \end{pmatrix}. \tag{1.1}$$

Here, the leading-block matrix G is a suitably-chosen approximation to H; it may either be prescribed **explicitly**, in which case a symmetric indefinite factorization of K_G will be formed using the GALAHAD package SLS, or **implicitly**, by requiring certain sub-blocks of G be zero. In the latter case, a factorization of K_G will be obtained implicitly (and more efficiently) without recourse to SLS. In particular, for implicit preconditioners, a reordering

$$K_G = P \begin{pmatrix} G_{11} & G_{21}^T & A_1^T \\ G_{21} & G_{22} & A_2^T \\ A_1 & A_2 & -C \end{pmatrix} P^T \tag{1.2}$$

involving a suitable permutation P will be found, for some invertible sub-block ("basis") A_1 of the columns of A; the selection and factorization of A_1 uses the GALAHAD package ULS. Once the preconditioner has been constructed, solutions to the preconditioning system

$$\begin{pmatrix} G & A^T \\ A & -C \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} \tag{1.3}$$

may be obtained by the package. Full advantage is taken of any zero coefficients in the matrices H, A and C.

ATTRIBUTES — Versions: GALAHAD_SBLS_single, GALAHAD_SBLS_double. **Uses:** GALAHAD_CLOCK, GALAHAD_SYMBOLS, GALAHAD_SPACE, GALAHAD_SMT, GALAHAD_QPT, GALAHAD_SLS, GALAHAD_ULS, GALAHAD_SPECFILE, **Date:** April 2006. **Origin:** H. S. Dollar and N. I. M. Gould, Rutherford Appleton Laboratory. **Language:** Fortran 95 + TR 15581 or Fortran 2003. **Parallelism:** Some options may use OpenMP and its runtime library.

2 HOW TO USE THE PACKAGE

The package is available with single, double and (if available) quadruple precision reals, and either 32-bit or 64-bit integers. Access to the 32-bit integer, single precision version requires the USE statement

```
USE GALAHAD_SBLS_single
```

with the obvious substitution GALAHAD_SBLS_double, GALAHAD_SBLS_quadruple, GALAHAD_SBLS_single_64, GALAHAD_SBLS_double_64 and GALAHAD_SBLS_quadruple_64 for the other variants.

If it is required to use more than one of the modules at the same time, the derived types SMT_type, QPT_problem_type, SBLS_time_type, SBLS_control_type, SBLS_inform_type and SBLS_data_type (§2.4) and the subroutines SBLS_initialize, SBLS_form_and_factorize, SBLS_solve, SBLS_terminate, (§2.5) and SBLS_read_specfile (§2.7) must be renamed on one of the USE statements.

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2.1 Matrix storage formats

Each of the input matrices $\boldsymbol{H}$, $\boldsymbol{A}$ and $\boldsymbol{C}$ may be stored in a variety of input formats.

2.1.1 Dense storage format

The matrix $\boldsymbol{A}$ is stored as a compact dense matrix by rows, that is, the values of the entries of each row in turn are stored in order within an appropriate real one-dimensional array. Component $n*(i-1)+j$ of the storage array `A%val` will hold the value a_{ij} for $i=1,\ldots,m$, $j=1,\ldots,n$. Since $\boldsymbol{H}$ and $\boldsymbol{C}$ are symmetric, only the lower triangular parts (that is the part h_{ij} for $1 \leq j \leq i \leq n$ and c_{ij} for $1 \leq j \leq i \leq m$) need be held. In these cases the lower triangle will be stored by rows, that is component $i*(i-1)/2+j$ of the storage array `H%val` will hold the value h_{ij} (and, by symmetry, h_{ji}) for $1 \leq j \leq i \leq n$. Similarly component $i*(i-1)/2+j$ of the storage array `C%val` will hold the value c_{ij} (and, by symmetry, c_{ji}) for $1 \leq j \leq i \leq m$.

2.1.2 Sparse co-ordinate storage format

Only the nonzero entries of the matrices are stored. For the l-th entry of $\boldsymbol{A}$, its row index i, column index j and value a_{ij} are stored in the l-th components of the integer arrays `A%row`, `A%col` and real array `A%val`, respectively. The order is unimportant, but the total number of entries `A%ne` is also required. The same scheme is applicable to $\boldsymbol{H}$ and $\boldsymbol{C}$ (thus, for $\boldsymbol{H}$, requiring integer arrays `H%row`, `H%col`, a real array `H%val` and an integer value `H%ne`), except that only the entries in the lower triangle need be stored.

2.1.3 Sparse row-wise storage format

Again only the nonzero entries are stored, but this time they are ordered so that those in row i appear directly before those in row $i+1$. For the i-th row of $\boldsymbol{A}$, the i-th component of a integer array `A%ptr` holds the position of the first entry in this row, while `A%ptr` $(m+1)$ holds the total number of entries plus one. The column indices j and values a_{ij} of the entries in the i-th row are stored in components $l=$ `A%ptr`(i), $\ldots$,`A%ptr` $(i+1)-1$ of the integer array `A%col`, and real array `A%val`, respectively. The same scheme is applicable to $\boldsymbol{H}$ and $\boldsymbol{C}$ (thus, for $\boldsymbol{H}$, requiring integer arrays `H%ptr`, `H%col`, and a real array `H%val`), except that only the entries in the lower triangle need be stored.

For sparse matrices, this scheme almost always requires less storage than its predecessor.

2.1.4 Diagonal storage format

If $\boldsymbol{H}$ is diagonal (i.e., $h_{ij}=0$ for all $1 \leq i \neq j \leq n$) only the diagonals entries h_{ii}, $1 \leq i \leq n$, need be stored, and the first n components of the array `H%val` may be used for the purpose. The same applies to $\boldsymbol{C}$, but there is no sensible equivalent for the non-square $\boldsymbol{A}$.

2.1.5 Scaled-identity storage format

If $\boldsymbol{H}$ is a scalar multiple h of the identity matrix, $h\boldsymbol{I}$, only the value h needs be stored, and the first component of the array `H%val` may be used for the purpose. The same applies to $\boldsymbol{C}$, but as before there is no sensible equivalent for the non-square $\boldsymbol{A}$.

2.1.6 Identity storage format

If $\boldsymbol{H}$ is the identity matrix, $\boldsymbol{I}$, no numerical data need be stored. The same applies to $\boldsymbol{C}$, but once again there is no sensible equivalent for the non-square $\boldsymbol{A}$.

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2.1.7 Zero storage format

Finally, if $\boldsymbol{H}$ is the zero matrix, $\mathbf{0}$, no numerical data need be stored. The same applies to both $\boldsymbol{A}$ and $\boldsymbol{C}$.

2.2 Real and integer kinds

We use the terms integer and real to refer to the fortran keywords `REAL(rp_)` and `INTEGER(ip_)`, where `rp_` and `ip_` are the relevant kind values for the real and integer types employed by the particular module in use. The former are equivalent to default `REAL` for the single precision versions, `DOUBLE PRECISION` for the double precision cases and quadruple-precision if 128-bit reals are available, and correspond to `rp_ = real32`, `rp_ = real64` and `rp_ = real128` respectively as defined by the fortran `iso_fortran_env` module. The latter are default (32-bit) and long (64-bit) integers, and correspond to `ip_ = int32` and `ip_ = int64`, respectively, again from the `iso_fortran_env` module.

2.3 Parallel usage

OpenMP may be used by the `GALAHAD_SBLS` package to provide parallelism for some solvers in shared memory environments. See the documentation for the GALAHAD package `SLS` for more details. To run in parallel, OpenMP must be enabled at compilation time by using the correct compiler flag (usually some variant of `-openmp`). The number of threads may be controlled at runtime by setting the environment variable `OMP_NUM_THREADS`.

MPI may also be used by the package to provide parallelism for some solvers in a distributed memory environment. To use this form of parallelism, MPI must be enabled at runtime by using the correct compiler flag (usually some variant of `-lmpi`). Although the MPI process will be started automatically when required, it should be stopped by the calling program once no further use of this form of parallelism is needed. Typically, this will be via statements of the form

```
CALL MPI_INITIALIZED( flag, ierr )
IF ( flag ) CALL MPI_FINALIZE( ierr )
```

The code may be compiled and run in serial mode.

2.4 The derived data types

Six derived data types are accessible from the package.

2.4.1 The derived data type for holding matrices

The derived data type `SMT_TYPE` is used to hold the matrices $\boldsymbol{H}$, $\boldsymbol{A}$ and $\boldsymbol{C}$. The components of `SMT_TYPE` used here are:

`m` is a scalar component of type `INTEGER(ip_)`, that holds the number of rows in the matrix.

`n` is a scalar component of type `INTEGER(ip_)`, that holds the number of columns in the matrix.

`type` is a rank-one allocatable array of type default `CHARACTER`, that is used to indicate the storage scheme used. If the dense storage scheme (see §2.1.1), is used, the first five components of `type` must contain the string `DENSE`. For the sparse co-ordinate scheme (see §2.1.2), the first ten components of `type` must contain the string `COORDINATE`, for the sparse row-wise storage scheme (see §2.1.3), the first fourteen components of `type` must contain the string `SPARSE_BY_ROWS`, for the diagonal storage scheme (see §2.1.4), the first eight components of `type` must contain the string `DIAGONAL`. for the scaled-identity storage scheme (see §2.1.5), the first fifteen components of `type` must contain the string `SCALED_IDENTITY`. and for the identity storage scheme (see §2.1.6), the first eight components of `type` must contain the string `IDENTITY`. It is also permissible to set the first four components of `type` to the either of the strings `ZERO` or `NONE` in the case of matrix $\boldsymbol{C}$ to indicate that $\boldsymbol{C} = 0$.

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For convenience, the procedure SMT_put may be used to allocate sufficient space and insert the required keyword into type. For example, if H is of derived type SMT_type and we wish to use the co-ordinate storage scheme, we may simply

```
CALL SMT_put( H%type, 'COORDINATE', istat )
```

See the documentation for the GALAHAD package SMT for further details on the use of SMT_put.

ne is a scalar variable of type INTEGER(ip_), that holds the number of matrix entries.

val is a rank-one allocatable array of type REAL(rp_) and dimension at least ne, that holds the values of the entries. Each pair of off-diagonal entries $h_{ij} = h_{ji}$ of a *symmetric* matrix $\boldsymbol{H}$ is represented as a single entry (see §2.1.1–2.1.3); the same applies to $\boldsymbol{C}$. Any duplicated entries that appear in the sparse co-ordinate or row-wise schemes will be summed. If the matrix is stored using the diagonal scheme (see §2.1.4), val should be of length n, and the value of the i-th diagonal stored in val(i). If the matrix is stored using the scaled-identity scheme (see §2.1.6), val(1) should be set to h.

row is a rank-one allocatable array of type INTEGER(ip_), and dimension at least ne, that may hold the row indices of the entries. (see §2.1.2).

col is a rank-one allocatable array of type INTEGER(ip_), and dimension at least ne, that may hold the column indices of the entries (see §2.1.2–2.1.3).

ptr is a rank-one allocatable array of type INTEGER(ip_), and dimension at least m + 1, that may hold the pointers to the first entry in each row (see §2.1.3).

2.4.2 The derived data type for holding control parameters

The derived data type SBLS_control_type is used to hold controlling data. Default values may be obtained by calling SBLS_initialize (see §2.5.1), while components may also be changed by calling GALAHAD_SBLS_read_spec (see §2.7.1). The components of SBLS_control_type are:

error is a scalar variable of type INTEGER(ip_), that holds the stream number for error messages. Printing of error messages in SBLS_solve and SBLS_terminate is suppressed if error ≤ 0. The default is error = 6.

out is a scalar variable of type INTEGER(ip_), that holds the stream number for informational messages. Printing of informational messages in SBLS_solve is suppressed if out < 0. The default is out = 6.

print_level is a scalar variable of type INTEGER(ip_), that is used to control the amount of informational output which is required. No informational output will occur if print_level ≤ 0. If print_level $= 1$, a single line of output will be produced for each iteration of the process. If print_level ≥ 2, this output will be increased to provide significant detail of each iteration. The default is print_level = 0.

new_h is a scalar variable of type INTEGER(ip_), that is used to indicate how $\boldsymbol{H}$ has changed (if at all) since the previous call to SBLS_form_and_factorize. Possible values are:

0 $\boldsymbol{H}$ is unchanged

1 the values in $\boldsymbol{H}$ have changed, but its nonzero structure is as before.

2 both the values and structure of $\boldsymbol{H}$ have changed.

The default is new_h = 2.

new_a is a scalar variable of type INTEGER(ip_), that is used to indicate how $\boldsymbol{A}$ has changed (if at all) since the previous call to SBLS_form_and_factorize. Possible values are:

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0 $\boldsymbol{A}$ is unchanged

1 the values in $\boldsymbol{A}$ have changed, but its nonzero structure is as before.

2 both the values and structure of $\boldsymbol{A}$ have changed.

The default is `new_a = 2`.

`new_c` is a scalar variable of type `INTEGER(ip_)`, that is used to indicate how $\boldsymbol{C}$ has changed (if at all) since the previous call to `SBLS_form_and_factorize`. Possible values are:

0 $\boldsymbol{C}$ is unchanged

1 the values in $\boldsymbol{C}$ have changed, but its nonzero structure is as before.

2 both the values and structure of $\boldsymbol{C}$ have changed.

The default is `new_c = 2`.

`preconditioner` is a scalar variable of type `INTEGER(ip_)`, that specifies which preconditioner to be used; positive values correspond to explicit-factorization preconditioners while negative values indicate implicit-factorization ones. Possible values are:

0 the preconditioner is chosen automatically on the basis of which option looks most likely to be the most efficient.

1 $\boldsymbol{G}$ is chosen to be the identity matrix.

2 $\boldsymbol{G}$ is chosen to be $\boldsymbol{H}$

3 $\boldsymbol{G}$ is chosen to be the diagonal matrix whose diagonals are the larger of those of $\boldsymbol{H}$ and a positive constant (see `min_diagonal` below).

4 $\boldsymbol{G}$ is chosen to be the band matrix of given semi-bandwidth whose entries coincide with those of $\boldsymbol{H}$ within the band. (see `semi_bandwidth` below).

5 $\boldsymbol{G}$ is chosen to be a diagonal matrix $\boldsymbol{D}$ specified by the user.

11 $\boldsymbol{G}$ is chosen so that $\boldsymbol{G}_{11} = 0$, $\boldsymbol{G}_{21} = 0$ and $\boldsymbol{G}_{22} = \boldsymbol{H}_{22}$.

12 $\boldsymbol{G}$ is chosen so that $\boldsymbol{G}_{11} = 0$, $\boldsymbol{G}_{21} = \boldsymbol{H}_{21}$ and $\boldsymbol{G}_{22} = \boldsymbol{H}_{22}$.

-1 for the special case when $\boldsymbol{C} = 0$, $\boldsymbol{G}$ is chosen so that $\boldsymbol{G}_{11} = 0$, $\boldsymbol{G}_{21} = 0$, $\boldsymbol{G}_{22}$ is the identity matrix, and the preconditioner is computed implicitly.

-2 for the special case when $\boldsymbol{C} = 0$, $\boldsymbol{G}$ is chosen so that $\boldsymbol{G}_{11} = 0$, $\boldsymbol{G}_{21} = 0$, $\boldsymbol{G}_{22} = \boldsymbol{H}_{22}$ and the preconditioner is computed implicitly.

Other values may be added in future. The default is `preconditioner = 0`.

`semi_bandwidth` is a scalar variable of type `INTEGER(ip_)`, that specifies the semi-bandwidth of the band preconditioner when `precon = 3`, if appropriate. The default is `semi_bandwidth = 5`.

`factorization` is a scalar variable of type `INTEGER(ip_)`, that specifies which factorization of the preconditioner should be used. Possible values are:

0 the factorization is chosen automatically on the basis of which option looks likely to be the most efficient.

1 if $\boldsymbol{G}$ is diagonal and non-singular, a Schur-complement factorization, involving factors of $\boldsymbol{G}$ and $\boldsymbol{A}\boldsymbol{G}^{-1}\boldsymbol{A}^T$, will be used. Otherwise an augmented-system factorization, involving factors of $\boldsymbol{K}_G$, will be used.

2 an augmented-system factorization, involving factors of $\boldsymbol{K}_G$, will be used.

3 a null-space factorization (see §4) will be used provided that $\boldsymbol{C} = \boldsymbol{0}$.

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The default is `factorization = 0`.

`max_col` is a scalar variable of type `INTEGER(ip_)`, that specifies the maximum number of nonzeros in a column of $\boldsymbol{A}$ which is permitted by the Schur-complement factorization. The default is `max_col = 35`.

`itref_max` is a scalar variable of type `INTEGER(ip_)`, that specifies the maximum number of iterative refinements allowed with each application of the preconditioner. The default is `itref_max = 1`.

`pivot_tol_for_basis` is a scalar variable of type default `REAL(rp_)`, that holds the relative pivot tolerance used by the package `ULS` when computing the non-singular basis matrix $\boldsymbol{A}_1$ for an implicit-factorization preconditioner. Since the calculation of a suitable basis is crucial, it is sensible to pick a larger value of `pivot_tol_for_basis` than the default value `relative_pivot_tolerance` provided by `ULS`. The default is `pivot_tol_for_basis = 0.5`.

`min_diagonal` is a scalar variable of type `REAL(rp_)`, that specifies the smallest permitted diagonal in $\boldsymbol{G}$ for some of the preconditioners provided. See `preconditioner` above. The default is `min_diagonal = 0.00001`.

`remove_dependencies` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if linear dependent rows from the second block equation $\boldsymbol{Ax} - \boldsymbol{Cy} = \boldsymbol{b}$ from (1.3) should be removed and `.FALSE.` otherwise. The default is `remove_dependencies = .TRUE.`.

`check_basis` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the basis matrix $\boldsymbol{A}_1$ constructed when using an implicit-factorization preconditioner or null-space factorization should be assessed for ill conditioning and corrected if necessary. If these precautions are not thought necessary, `check_basis` should be set `.FALSE.`. The default is `check_basis = .TRUE.`.

`find_basis_by_transpose` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the invertible sub-block $\boldsymbol{A}_1$ of the columns of $\boldsymbol{A}$ is computed by analysing the transpose of $\boldsymbol{A}$ and `.FALSE.` if the analysis is based on $\boldsymbol{A}$ itself. Generally an analysis based on the transpose is faster. The default is `find_basis_by_transpose = .TRUE.`.

`affine` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the component $\boldsymbol{b}$ of the right-hand side of the required system (1.3) is zero, and `.FALSE.` otherwise. Computational savings are possible when $\boldsymbol{b} = 0$. The default is `affine = .FALSE.`.

`perturb_to_make_definite` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the user wants to guarantee that the computed preconditioner is suitable by boosting the diagonal of the requested $\boldsymbol{G}$ and `.FALSE.` otherwise. The default is `perturb_to_make_definite = .TRUE.`.

`get_norm_residual` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the user wishes the package to return the value of the norm of the residuals for the computed solution when applying the preconditioner and `.FALSE.` otherwise. The default is `get_norm_residual = .FALSE.`.

`space_critical` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if space is critical when allocating arrays and `.FALSE.` otherwise. The package may run faster if `space_critical` is `.FALSE.` but at the possible expense of a larger storage requirement. The default is `space_critical = .FALSE.`.

`deallocate_error_fatal` is a scalar variable of type default `LOGICAL`, that must be set `.TRUE.` if the user wishes to terminate execution if a deallocation fails, and `.FALSE.` if an attempt to continue will be made. The default is `deallocate_error_fatal = .FALSE.`.

`symmetric_linear_solver` is a scalar variable of type default `CHARACTER` and length 30, that specifies the external package to be used to solve any general symmetric linear systems that might arise. Possible choices are `'sils'`, `'ma27'`, `'ma57'`, `'ma77'`, `'ma86'`, `'ma97'`, `'ssids'`, `'pardiso'`, `'wsmp'`, `'sytr'`, although only `'sils'`, `'sytr'` and, for OMP 4.0-compliant compilers, `'ssids'` are installed by default. See the documentation for the GALAHAD package `SLS` for further details. The default is `symmetric_linear_solver = 'ssids'`.

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`definite_linear_solver` is a scalar variable of type default `CHARACTER` and length 30, that specifies the external package to be used to solve any symmetric positive-definite linear system that might arise. Possible choices are `'sils'`, `'ma27'`, `'ma57'`, `'ma77'`, `'ma86'`, `'ma87'`, `'ma97'`, `'ssids'`, `'pardiso'`, `'mkl_pardiso'`, `'wsmp'`, `'potr'` and `'pbtr'`, although only `'sils'`, `'potr'`, `'pbtr'` and, for OMP 4.0-compliant compilers, `'ssids'` are installed by default. See the documentation for the GALAHAD package `SLS` for further details. The default is `definite_linear_solver = 'ssids'`.

`unsymmetric_linear_solver` is a scalar variable of type default `CHARACTER` and length 30, that specifies the external package to be used to solve any unsymmetric linear systems that might arise. Possible choices are `'gls'`, `'ma28'` and `'ma48'`, although only `'gls'` is installed by default. See the documentation for the GALAHAD package `ULS` for further details. The default is `unsymmetric_linear_solver = 'gls'`.

`prefix` is a scalar variable of type default `CHARACTER` and length 30, that may be used to provide a user-selected character string to preface every line of printed output. Specifically, each line of output will be prefaced by the string `prefix(2:LEN(TRIM(prefix))-1)`, thus ignoring the first and last non-null components of the supplied string. If the user does not want to preface lines by such a string, they may use the default `prefix = ""`.

`SLS_control` is a scalar variable argument of type `SLS_control_type` that is used to pass control options to external packages used to solve any symmetric linear systems that might arise. See the documentation for the GALAHAD package `SLS` for further details. In particular, default values are as for `SLS`.

`ULS_control` is a scalar variable argument of type `ULS_control_type` that is used to pass control options to external packages used to solve any unsymmetric linear systems that might arise. See the documentation for the GALAHAD package `ULS` for further details. In particular, default values are as for `ULS`.

2.4.3 The derived data type for holding timing information

The derived data type `SBLS_time_type` is used to hold elapsed CPU and system clock times for the various parts of the calculation. The components of `SBLS_time_type` are:

`total` is a scalar variable of type `REAL(rp_)`, that gives the total CPU time (in seconds) spent in the package.

`assemble` is a scalar variable of type `REAL(rp_)`, that gives the CPU time spent forming the preconditioner.

`factorize` is a scalar variable of type `REAL(rp_)`, that gives the CPU time spent factorizing the preconditioner.

`solve` is a scalar variable of type `REAL(rp_)`, that gives the CPU time spent applying the preconditioner.

`update` is a scalar variable of type `REAL(rp_)`, that gives the CPU time spent updating the factorization.

`clock_total` is a scalar variable of type `REAL(rp_)`, that gives the total elapsed system clock time (in seconds) spent in the package.

`clock_assemble` is a scalar variable of type `REAL(rp_)`, that gives the elapsed system clock time spent forming the preconditioner.

`clock_factorize` is a scalar variable of type `REAL(rp_)`, that gives the elapsed system clock time spent factorizing the preconditioner.

`clock_solve` is a scalar variable of type `REAL(rp_)`, that gives the elapsed system clock time spent applying the preconditioner.

`clock_update` is a scalar variable of type `REAL(rp_)`, that gives the elapsed system clock time spent updating the factorization.

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2.4.4 The derived data type for holding informational parameters

The derived data type SBLS_inform_type is used to hold parameters that give information about the progress and needs of the algorithm. The components of SBLS_inform_type are:

status is a scalar variable of type INTEGER(ip_), that gives the exit status of the algorithm. See §2.6 for details.

alloc_status is a scalar variable of type INTEGER(ip_), that gives the status of the last attempted array allocation or deallocation. This will be 0 if status = 0.

bad_alloc is a scalar variable of type default CHARACTER and length 80, that gives the name of the last internal array for which there were allocation or deallocation errors. This will be the null string if status = 0.

factorization_integer is a scalar variable of type INTEGER(int64), that reports the number of integers required to hold the factorization.

factorization_real is a scalar variable of type INTEGER(int64), that reports the number of reals required to hold the factorization.

preconditioner is a scalar variable of type INTEGER(ip_), that indicates the preconditioner method used. The range of values returned corresponds to those requested in control%preconditioner, excepting that the requested value may have been altered to a more appropriate one during the factorization. In particular, if the automatic choice control%preconditioner = 0 is requested, preconditioner reports the actual choice made.

factorization is a scalar variable of type INTEGER(ip_), that indicates the factorization method used. The range of values returned corresponds to those requested in control%factorization, excepting that the requested value may have been altered to a more appropriate one during the factorization. In particular, if the automatic choice control%factorization = 0 is requested, factorization reports the actual choice made.

rank is a scalar variable of type INTEGER(ip_), that gives the computed rank of $\boldsymbol{A}$.

rank_def is a scalar variable of type default LOGICAL, that has the value .TRUE. if SBLS_form_and_factorize believes that $\boldsymbol{A}$ is rank defficient, and .FALSE. otherwise

perturbed is a scalar variable of type default LOGICAL, that has the value .TRUE. if and only if the original choice of $\boldsymbol{G}$ has been perturbed to ensure that $\boldsymbol{K}_G$ is an appropriate preconditioner. This will only happen if control%perturb_to_make_definite has been set .TRUE..

norm_residual is a scalar variable of type REAL(rp_), that holds the infinity norm of the residual of the system (1.3) after a call to SBLS_solve if control%get_norm_residual has been set .TRUE.. Otherwise it will have the value -1.0.

time is a scalar variable of type SBLS_time_type whose components are used to hold elapsed CPU and system clock times (in seconds) for the various parts of the calculation (see Section 2.4.3).

SLS_inform is a scalar variable argument of type SLS_inform_type that is used to pass information concerning the progress of the external packages used to solve any symmetric linear systems that might arise. See the documentation for the GALAHAD package SLS for further details.

ULS_inform is a scalar variable argument of type ULS_inform_type that is used to pass information concerning the progress of the external packages used to solve any unsymmetric linear systems that might arise. See the documentation for the GALAHAD package ULS for further details.

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2.4.5 The derived data type for holding problem data

The derived data types SBLS_explicit_factor_type, SBLS_implicit_factor_type and SBLS_data_type are used to hold all the data for the problem and the factors of its preconditioners between calls of SBLS procedures. This data should be preserved, untouched, from the initial call to SBLS_initialize to the final call to SBLS_terminate.

2.5 Argument lists and calling sequences

There are four procedures for user calls (see §2.7 for further features):

1. The subroutine SBLS_initialize is used to set default values, and initialize private data, before solving one or more problems with the same sparsity and bound structure.
2. The subroutine SBLS_form_and_factorize is called to form and factorize the preconditioner.
3. The subroutine SBLS_solve is called to apply the preconditioner, that is to solve a linear system of the form (1.3).
4. The subroutine SBLS_terminate is provided to allow the user to automatically deallocate array components of the private data, allocated by SBLS_form_and_factorize at the end of the solution process.

We use square brackets [ ] to indicate OPTIONALarguments.

2.5.1 The initialization subroutine

Default values are provided as follows:

```
CALL SBLS_initialize( data, control, inform )
```

data is a scalar INTENT(INOUT) argument of type SBLS_data_type (see §2.4.5). It is used to hold data about the problem being solved.

control is a scalar INTENT(OUT) argument of type SBLS_control_type (see §2.4.2). On exit, control contains default values for the components as described in §2.4.2. These values should only be changed after calling SBLS_initialize.

inform is a scalar INTENT(OUT) argument of type SBLS_inform_type (see Section 2.4.4). A successful call to SBLS_initialize is indicated when the component status has the value 0. For other return values of status, see Section 2.6.

2.5.2 The subroutine for forming and factorizing the preconditioner

A preconditioner of the form (1.1) is formed and factorized as follows:

```
CALL SBLS_form_and_factorize( n, m, H, A, C, data, control, inform[, D] )
```

n is a scalar INTENT(IN) argument of type INTEGER(ip_) that specifies the number of rows of $\boldsymbol{H}$ (and columns of $\boldsymbol{A}$).

m is a scalar INTENT(IN) argument of type INTEGER(ip_) that specifies the number of rows of $\boldsymbol{A}$ and $\boldsymbol{C}$.

H is a scalar INTENT(IN) argument of type SMT_type whose components must be set to specify the data defining the matrix $\boldsymbol{H}$ (see §2.4.1).

A is a scalar INTENT(IN) argument of type SMT_type whose components must be set to specify the data defining the matrix $\boldsymbol{A}$ (see §2.4.1).

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C is a scalar INTENT(IN) argument of type SMT_type whose components must be set to specify the data defining the matrix $\boldsymbol{C}$ (see §2.4.1).

data is a scalar INTENT(INOUT) argument of type SBLS_data_type (see §2.4.5). It is used to hold data about the problem being solved. It must not have been altered **by the user** since the last call to SBLS_initialize.

control is a scalar INTENT(IN) argument of type SBLS_control_type (see §2.4.2). Default values may be assigned by calling SBLS_initialize prior to the first call to SBLS_solve.

inform is a scalar INTENT(OUT) argument of type SBLS_inform_type (see §2.4.4). A successful call to SBLS_solve is indicated when the component status has the value 0. For other return values of status, see §2.6.

D is a rank-one OPTIONAL INTENT(IN) argument of type REAL(rp_) and length at least n, whose i-th component give the value of the i-th diagonal entry of the matrix $\boldsymbol{D}$ used when control%preconditioner = 5 (see §2.4.2). D need not be PRESENT for other values of control%preconditioner.

2.5.3 The subroutine for applying the preconditioner

The preconditioner may be applied to solve a system of the form (1.3) as follows:

```
CALL SBLS_solve( n, m, A, C, data, control, inform, SOL )
```

Components n, m, A, C, data and control are exactly as described for SBLS_form_and_factorize and must not have been altered in the interim.

inform is a scalar INTENT(OUT) argument of type SBLS_inform_type (see §2.4.4), that should be passed unaltered since the last call to SBLS_form_and_factorize or SBLS_solve. A successful call to SBLS_solve is indicated when the component status has the value 0. For other return values of status, see §2.6.

SOL is a rank-one INTENT(INOUT) array of type default REAL and length at least n+m, that must be set on entry to hold the composite vector $(a^T \;\; b^T)^T$. In particular SOL(i), $i = 1, \ldots$ n should be set to a_i, and SOL(n+j), $j = 1, \ldots,$ m should be set to b_j. On successful exit, SOL will contain the solution $(x^T \;\; y^T)^T$ to (1.3), that is SOL(i), $i = 1, \ldots,$ n will give x_i, and SOL(n+j), $j = 1, \ldots,$ m will contain y_j.

2.5.4 The termination subroutine

All previously allocated arrays are deallocated as follows:

```
CALL SBLS_terminate( data, control, inform )
```

data is a scalar INTENT(INOUT) argument of type SBLS_data_type exactly as for SBLS_solve, which must not have been altered **by the user** since the last call to SBLS_initialize. On exit, array components will have been deallocated.

control is a scalar INTENT(IN) argument of type SBLS_control_type exactly as for SBLS_solve.

inform is a scalar INTENT(OUT) argument of type SBLS_inform_type exactly as for SBLS_solve. Only the component status will be set on exit, and a successful call to SBLS_terminate is indicated when this component status has the value 0. For other return values of status, see §2.6.

2.6 Warning and error messages

A negative value of inform%status on exit from SBLS_form_and_factorize, SBLS_solve or SBLS_terminate indicates that an error has occurred. No further calls should be made until the error has been corrected. Possible values are:

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-1. An allocation error occurred. A message indicating the offending array is written on unit control%error, and the returned allocation status and a string containing the name of the offending array are held in inform%alloc_status and inform%bad_alloc respectively.

-2. A deallocation error occurred. A message indicating the offending array is written on unit control%error and the returned allocation status and a string containing the name of the offending array are held in inform%alloc_status and inform%bad_alloc respectively.

-3. One of the restrictions prob%n > 0 or prob%m ≥ 0 or requirements that prob%A_type, prob%H_type and prob%C_type contain its relevant string 'DENSE', 'COORDINATE', 'SPARSE_BY_ROWS', 'DIAGONAL', 'SCALED_IDENTITY', 'IDENTITY', 'ZERO' or 'NONE' has been violated.

-9. An error was reported by SLS_analyse. The return status from SLS_analyse is given in inform%SLS_inform%status. See the documentation for the GALAHAD package SLS for further details.

-10. An error was reported by SLS_factorize. The return status from SLS_factorize is given in inform%SLS_inform%status. See the documentation for the GALAHAD package SLS for further details.

-11. An error was reported by SLS_solve. The return status from SLS_solve is given in inform%SLS_inform%status. See the documentation for the GALAHAD package SLS for further details.

-13. An error was reported by ULS_factorize. The return status from ULS_factorize is given in inform%uls_factorize_status. See the documentation for the GALAHAD package ULS for further details.

-14. An error was reported by ULS_solve. The return status from ULS_solve is given in inform%uls_solve_status. See the documentation for the GALAHAD package ULS for further details.

-15. The computed preconditioner is singular and is thus unsuitable.

-20. The computed preconditioner has the wrong inertia and is thus unsuitable.

-24. An error was reported by SORT_reorder_by_rows. The return status from SORT_reorder_by_rows is given in inform%sort_status. See the documentation for the GALAHAD package SORT for further details.

-26. The requested linear equation solver is not available.

A positive value of inform%status on exit from SBLS_form_and_factorize warns of unexpected behaviour. A possible values is:

1. The matrx $\boldsymbol{A}$ is rank defficient.

2.7 Further features

In this section, we describe an alternative means of setting control parameters, that is components of the variable control of type SBLS_control_type (see §2.4.2), by reading an appropriate data specification file using the subroutine SBLS_read_specfile. This facility is useful as it allows a user to change SBLS control parameters without editing and recompiling programs that call SBLS.

A specification file, or specfile, is a data file containing a number of "specification commands". Each command occurs on a separate line, and comprises a "keyword", which is a string (in a close-to-natural language) used to identify a control parameter, and an (optional) "value", which defines the value to be assigned to the given control parameter. All keywords and values are case insensitive, keywords may be preceded by one or more blanks but values must not contain blanks, and each value must be separated from its keyword by at least one blank. Values must not contain more than 30 characters, and each line of the specfile is limited to 80 characters, including the blanks separating keyword and value.

The portion of the specification file used by SBLS_read_specfile must start with a "BEGIN SBLS" command and end with an "END" command. The syntax of the specfile is thus defined as follows:

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```
( .. lines ignored by SBLS_read_specfile .. )
  BEGIN SBLS
     keyword    value
     .......    .....
     keyword    value
  END
( .. lines ignored by SBLS_read_specfile .. )
```

where keyword and value are two strings separated by (at least) one blank. The "BEGIN SBLS" and "END" delimiter command lines may contain additional (trailing) strings so long as such strings are separated by one or more blanks, so that lines such as

```
BEGIN SBLS SPECIFICATION
```

and

```
END SBLS SPECIFICATION
```

are acceptable. Furthermore, between the "BEGIN SBLS" and "END" delimiters, specification commands may occur in any order. Blank lines and lines whose first non-blank character is ! or * are ignored. The content of a line after a ! or * character is also ignored (as is the ! or * character itself). This provides an easy manner to "comment out" some specification commands, or to comment specific values of certain control parameters.

The value of a control parameters may be of three different types, namely integer, logical or real. Integer and real values may be expressed in any relevant Fortran integer and floating-point formats (respectively). Permitted values for logical parameters are "ON", "TRUE", ".TRUE.", "T", "YES", "Y", or "OFF", "NO", "N", "FALSE", ".FALSE." and "F". Empty values are also allowed for logical control parameters, and are interpreted as "TRUE".

The specification file must be open for input when SBLS_read_specfile is called, and the associated device number passed to the routine in device (see below). Note that the corresponding file is REWINDed, which makes it possible to combine the specifications for more than one program/routine. For the same reason, the file is not closed by SBLS_read_specfile.

Control parameters corresponding to the components SLS_control and ULS_control may be changed by including additional sections enclosed by "BEGIN SLS" and "END SLS", and "BEGIN ULS" and "END ULS", respectively. See the specification sheets for the packages GALAHAD_SLS and GALAHAD_ULS for further details.

2.7.1 To read control parameters from a specification file

Control parameters may be read from a file as follows:

```
CALL SBLS_read_specfile( control, device )
```

control is a scalar INTENT(INOUT) argument of type SBLS_control_type (see §2.4.2). Default values should have already been set, perhaps by calling SBLS_initialize. On exit, individual components of control may have been changed according to the commands found in the specfile. Specfile commands and the component (see §2.4.2) of control that each affects are given in Table 2.1.

device is a scalar INTENT(IN) argument of type INTEGER(ip_), that must be set to the unit number on which the specfile has been opened. If device is not open, control will not be altered and execution will continue, but an error message will be printed on unit control%error.

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command	component of control	value type
error-printout-device	%error	integer
printout-device	%out	integer
print-level	%print_level	integer
maximum-refinements	%itref_max	integer
preconditioner-used	%preconditioner	integer
semi-bandwidth-for-band-preconditioner	%semi_bandwidth	integer
factorization-used	%factorization	integer
maximum-column-nonzeros-in-schur-complement	%max_col	integer
has-a-changed	%new_a	integer
has-h-changed	%new_h	integer
has-c-changed	%new_c	integer
minimum-diagonal	%min_diagonal	real
pivot-tolerance-used-for-basis	%pivot_tol_for_basis	real
find-basis-by-transpose	%find_basis_by_transpose	logical
remove-linear-dependencies	%remove_dependencies	logical
affine-constraints	%affine	logical
check-for-reliable-basis	%check_basis	logical
perturb-to-make-+ve-definite	%perturb_to_make_definite	logical
get-norm-residual	%get_norm_residual	logical
space-critical	%space_critical	logical
deallocate-error-fatal	%deallocate_error_fatal	logical
symmetric-linear-equation-solver	%symmetric_linear_solver	character
definite-linear-equation-solver	%definite_linear_solver	character
unsymmetric-linear-equation-solver	%unsymmetric_linear_solver	character
output-line-prefix	%prefix	character

Table 2.1: Specfile commands and associated components of control.

2.8 Information printed

If control%print_level is positive, information about the progress of the algorithm will be printed on unit control%out. If control%print_level $= 1$, statistics concerning the factorization, as well as warning and error messages will be reported. If control%print_level $= 2$, additional information about the progress of the factorization and the solution phases will be given. If control%print_level > 2, debug information, of little interest to the general user, may be returned.

3 GENERAL INFORMATION

Use of common: None.

Workspace: Provided automatically by the module.

Other routines called directly: None.

Other modules used directly: SBLS_solve calls the GALAHAD packages GALAHAD_CLOCK, GALAHAD_SYMBOLS, GALAHAD_SPACE, GALAHAD_SMT, GALAHAD_QPT, GALAHAD_SLS, GALAHAD_ULS and GALAHAD_SPECFILE,

Input/output: Output is under control of the arguments control%error, control%out and control%print_level.

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Restrictions: `prob%n` > 0, `prob%m` ≥ 0, `prob%H_type`, `prob%A_type` and `prob%C_type` $\in$ {`'DENSE'`, `'COORDINATE'`, `'SPARSE_BY_ROWS'`, `'DIAGONAL'`, `'SCALED_IDENTITY'`, `'IDENTITY'` }, `'ZERO'`, `'NONE'`.

Portability: ISO Fortran 95 + TR 15581 or Fortran 2003. The package is thread-safe.

4 METHOD

The method used depends on whether an explicit or implicit factorization is required. In the explicit case, the package is really little more than a wrapper for the symmetric, indefinite linear solver `SLS` in which the system matrix $\boldsymbol{K}_G$ is assembled from its constituents $\boldsymbol{A}$, $\boldsymbol{C}$ and whichever $\boldsymbol{G}$ is requested by the user. Implicit-factorization preconditioners are more involved, and there is a large variety of different possibilities. The essential ideas are described in detail in

H. S. Dollar, N. I. M. Gould and A. J. Wathen. "On implicit-factorization constraint preconditioners". In Large Scale Nonlinear Optimization (G. Di Pillo and M. Roma, eds.) Springer Series on Nonconvex Optimization and Its Applications, Vol. 83, Springer Verlag (2006) 61–82

and

H. S. Dollar, N. I. M. Gould, W. H. A. Schilders and A. J. Wathen "On iterative methods and implicit-factorization preconditioners for regularized saddle-point systems". SIAM Journal on Matrix Analysis and Applications, **28(1)** (2006) 170–189.

The range-space factorization is based upon the decomposition

$$\boldsymbol{K}_G = \begin{pmatrix} \boldsymbol{G} & \boldsymbol{0} \\ \boldsymbol{A} & \boldsymbol{I} \end{pmatrix} \begin{pmatrix} \boldsymbol{G}^{-1} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{S} \end{pmatrix} \begin{pmatrix} \boldsymbol{G} & \boldsymbol{A}^T \\ \boldsymbol{0} & \boldsymbol{I} \end{pmatrix},$$

where the "Schur complement" $\boldsymbol{S} = \boldsymbol{C} + \boldsymbol{A}\boldsymbol{G}^{-1}\boldsymbol{A}^T$. Such a method requires that $\boldsymbol{S}$ is easily invertible. This is often the case when $\boldsymbol{G}$ is a diagonal matrix, in which case $\boldsymbol{S}$ is frequently sparse, or when $m \ll n$ in which case $\boldsymbol{S}$ is small and a dense Cholesky factorization may be used.

When $\boldsymbol{C} = 0$, the null-space factorization is based upon the decomposition

$$\boldsymbol{K}_G = \boldsymbol{P} \begin{pmatrix} \boldsymbol{G}_{11} & \boldsymbol{0} & \boldsymbol{I} \\ \boldsymbol{G}_{21} & \boldsymbol{I} & \boldsymbol{A}_2^T\boldsymbol{A}_1^{-T} \\ \boldsymbol{A}_1 & \boldsymbol{0} & \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I} \\ \boldsymbol{0} & \boldsymbol{R} & \boldsymbol{0} \\ \boldsymbol{I} & \boldsymbol{0} & -\boldsymbol{G}_{11} \end{pmatrix} \begin{pmatrix} \boldsymbol{G}_{11} & \boldsymbol{G}_{21}^T & \boldsymbol{A}_1^T \\ \boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{I} & \boldsymbol{A}_1^{-1}\boldsymbol{A}_2 & \boldsymbol{0} \end{pmatrix} \boldsymbol{P}^T,$$

where the "reduced Hessian"

$$\boldsymbol{R} = (-\boldsymbol{A}_2^T\boldsymbol{A}_1^{-T} \;\; \boldsymbol{I}) \begin{pmatrix} \boldsymbol{G}_{11} & \boldsymbol{G}_{21}^T \\ \boldsymbol{G}_{21} & \boldsymbol{G}_{22} \end{pmatrix} \begin{pmatrix} -\boldsymbol{A}_1^{-1}\boldsymbol{A}_2 \\ \boldsymbol{I} \end{pmatrix}$$

and $\boldsymbol{P}$ is a suitably-chosen permutation for which $\boldsymbol{A}_1$ is invertible. The method is most useful when $m \approx n$ as then the dimension of $\boldsymbol{R}$ is small and a dense Cholesky factorization may be used.

5 EXAMPLE OF USE

Suppose we wish to solve the linear system (1.3) with matrix data

$$\boldsymbol{H} = \begin{pmatrix} 1 & & 4 \\ & 2 & \\ 4 & & 3 \end{pmatrix}, \; \boldsymbol{A} = \begin{pmatrix} 2 & 1 & \\ & 1 & 1 \end{pmatrix} \text{ and } \boldsymbol{C} = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}$$

and right-hand sides

$$\boldsymbol{a} = \begin{pmatrix} 7 \\ 4 \\ 8 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

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Then storing the matrices in sparse co-ordinate format, we may use the following code:

```
! THIS VERSION: GALAHAD 2.1 - 22/03/2007 AT 09:00 GMT.
   PROGRAM GALAHAD_SBLS_EXAMPLE
   USE GALAHAD_SBLS_double         ! double precision version
   IMPLICIT NONE
   INTEGER, PARAMETER :: wp = KIND( 1.0D+0 ) ! set precision
   TYPE ( SMT_type ) :: H, A, C
   REAL ( KIND = wp ), ALLOCATABLE, DIMENSION( : ) :: SOL
   TYPE ( SBLS_data_type ) :: data
   TYPE ( SBLS_control_type ) :: control
   TYPE ( SBLS_inform_type ) :: inform
   INTEGER :: s
   INTEGER :: n = 3, m = 2, h_ne = 4, a_ne = 4, c_ne = 1
! start problem data
   ALLOCATE( SOL( n + m ) )
   SOL( 1 : n ) = (/ 7.0_wp, 4.0_wp, 8.0_wp /)  ! RHS a
   SOL( n + 1 : n + m ) = (/ 2.0_wp, 1.0_wp /)  ! RHS b
! sparse co-ordinate storage format
   CALL SMT_put( H%type, 'COORDINATE', s )  ! Specify co-ordinate
   CALL SMT_put( A%type, 'COORDINATE', s )  ! storage for H, A and C
   CALL SMT_put( C%type, 'COORDINATE', s )
   ALLOCATE( H%val( h_ne ), H%row( h_ne ), H%col( h_ne ) )
   ALLOCATE( A%val( a_ne ), A%row( a_ne ), A%col( a_ne ) )
   ALLOCATE( C%val( c_ne ), C%row( c_ne ), C%col( c_ne ) )
   H%val = (/ 1.0_wp, 2.0_wp, 3.0_wp, 4.0_wp /) ! matrix H
   H%row = (/ 1, 2, 3, 3 /)                     ! NB lower triangle
   H%col = (/ 1, 2, 3, 1 /) ; H%ne = h_ne
   A%val = (/ 2.0_wp, 1.0_wp, 1.0_wp, 1.0_wp /) ! matrix A
   A%row = (/ 1, 1, 2, 2 /)
   A%col = (/ 1, 2, 2, 3 /) ; A%ne = a_ne
   C%val = (/ 1.0_wp /) ! matrix C
   C%row = (/ 2 /) ! NB lower triangle
   C%col = (/ 1 /) ; C%ne = c_ne
! problem data complete
   CALL SBLS_initialize( data, control, inform ) ! Initialize control parameters
   control%preconditioner = 2                    ! Exact factorization
! factorize matrix
   CALL SBLS_form_and_factorize( n, m, H, A, C, data, control, inform )
   IF ( inform%status < 0 ) THEN                 ! Unsuccessful call
     WRITE( 6, "( ' SBLS_form_and_factorize exit status = ', I0 ) " )          &
       inform%status
     STOP
   END IF
! solve system
   CALL SBLS_solve( n, m, A, C, data, control, inform, SOL )
   IF ( inform%status == 0 ) THEN                ! Successful return
     WRITE( 6, "( ' SBLS: Solution = ', /, ( 5ES12.4 ) )" ) SOL
   ELSE                                          !  Error returns
     WRITE( 6, "( ' SBLS_solve exit status = ', I6 ) " ) inform%status
   END IF
   CALL SBLS_terminate( data, control, inform )  ! delete internal workspace
   END PROGRAM GALAHAD_SBLS_EXAMPLE
```

This produces the following output:

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```
 SBLS: Solution =
  1.0000E+00  1.0000E+00  1.0000E+00  1.0000E+00  1.0000E+00
```

The same problem may be solved holding the data in a sparse row-wise storage format by replacing the lines

```
!  sparse co-ordinate storage format
...
! problem data complete
```

by

```
! sparse row-wise storage format
   CALL SMT_put( H%type, 'SPARSE_BY_ROWS', s )  ! Specify sparse-by-rows
   CALL SMT_put( A%type, 'SPARSE_BY_ROWS', s )  ! storage for H, A and C
   CALL SMT_put( C%type, 'SPARSE_BY_ROWS', s )
   ALLOCATE( H%val( h_ne ), H%col( h_ne ), H%ptr( n + 1 ) )
   ALLOCATE( A%val( a_ne ), A%col( a_ne ), A%ptr( m + 1 ) )
   ALLOCATE( C%val( c_ne ), C%col( c_ne ), C%ptr( m + 1 ) )
   H%val = (/ 1.0_wp, 2.0_wp, 3.0_wp, 4.0_wp /) ! matrix H
   H%col = (/ 1, 2, 3, 1 /)                     ! NB lower triangular
   H%ptr = (/ 1, 2, 3, 5 /)                     ! Set row pointers
   A%val = (/ 2.0_wp, 1.0_wp, 1.0_wp, 1.0_wp /) ! matrix A
   A%col = (/ 1, 2, 2, 3 /)
   A%ptr = (/ 1, 3, 5 /)                        ! Set row pointers
   C%val = (/ 1.0_wp /)                         ! matrix C
   C%col = (/ 1 /)                              ! NB lower triangular
   C%ptr = (/ 1, 1, 2 /)                        ! Set row pointers
! problem data complete
```

or using a dense storage format with the replacement lines

```
! dense storage format
   CALL SMT_put( H%type, 'DENSE', s )  ! Specify dense
   CALL SMT_put( A%type, 'DENSE', s )  ! storage for H, A and C
   CALL SMT_put( C%type, 'DENSE', s )
   ALLOCATE( H%val( n * ( n + 1 ) / 2 ) )
   ALLOCATE( A%val( n * m ) )
   ALLOCATE( C%val( m * ( m + 1 ) / 2 ) )
   H%val = (/ 1.0_wp, 0.0_wp, 2.0_wp, 4.0_wp, 0.0_wp, 3.0_wp /) ! H
   A%val = (/ 2.0_wp, 1.0_wp, 0.0_wp, 0.0_wp, 1.0_wp, 1.0_wp /) ! A
   C%val = (/ 0.0_wp, 1.0_wp, 0.0_wp /)                         ! C
! problem data complete
```

respectively.

If instead $\boldsymbol{H}$ had been the diagonal matrix

$$\boldsymbol{H} = \begin{pmatrix} 1 & & \\ & 0 & \\ & & 3 \end{pmatrix}$$

but the other data is as before, the diagonal storage scheme might be used for $\boldsymbol{H}$, and in this case we would instead

```
   CALL SMT_put( prob%H%type, 'DIAGONAL', s )  ! Specify dense storage for H
   ALLOCATE( p%H%val( n ) )
   p%H%val = (/ 1.0_wp, 0.0_wp, 3.0_wp /) ! Hessian values
```

Notice here that zero diagonal entries are stored.

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